Dmitrii Nikolaev, Ph.D.

Senior ML Engineer | LLM Inference Optimization

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PROFILE

AI Engineer with a PhD in Applied Mathematics and 7+ years designing, deploying, and operating ML services and APIs in production. At AMD, I own training and inference performance for enterprise customers running on AMD and NVIDIA hardware, from initial benchmarks to production. My recent benchmark work increased Qwen-Image training throughput 16x; my production inference changes raised concurrent-request capacity by 55% while sustaining over 40,000 tokens/sec per 8-GPU node. Earlier I built end-to-end LLM evaluation, RAG search, and diffusion fine-tuning systems at Silo AI, multi-agent RL and edge AI systems at Skoltech, onboarded seven engineers, supervised eight graduated students on applied ML projects.

SKILLS

- **Distributed training:** PyTorch, DeepSpeed, Megatron-LM, FSDP, ZeRO, tensor, pipeline, context, sequence, and expert parallelism, fixed- and mixed-precision training, LoRA, transformers, diffusion.
- **Inference systems:** vLLM, SGLang, torch.compile performance optimization, KV-cache management, continuous batching, quantization, multi-GPU serving, latency and throughput benchmarking.
- **Accelerator performance:** CUDA, ROCm, C++, GPU/NPU profiling, Linux performance tooling, GEMM kernel debugging and tuning, roofline analysis, memory and compute bottleneck analysis.
- **Infrastructure and services:** Python, Go, REST APIs, FastAPI, Docker, Kubernetes, Helm, CI/CD, Prometheus, Grafana, AWS, Azure; production deployment, monitoring, and service operations.

EXPERIENCE

AMD **Helsinki, Finland**

Semiconductors and AI compute, joined via the Silo AI acquisition

Senior AI Engineer *Aug 2024 – Present*

- **Distributed training.** For a global electronics manufacturer, pinpointed a DeepSpeed 0.18.5 data/pipeline-parallel mapping failure in Qwen-Image 2.1 training (20B, BF16) on AMD Instinct MI300X, proved the issue was in DeepSpeed rather than AMD hardware, redesigned the pipeline parallelism layout. Increased throughput 16x, from 2 to 32 images/sec per 8-GPU node, cut compute time by 94%.
- **GPU performance.** For a global electronics manufacturer, built an automated benchmark suite and trade-off report comparing NVIDIA H100 and AMD Instinct MI300X for Llama 3.1 and Qwen3 (8B, BF16). Applied roofline analysis, evaluated parallelism and sharding strategies, and retuned GEMM kernels; optimized configurations delivered up to 21% higher throughput, increased Model FLOPs Utilization from 38% to 46%, and indicated ∼17% lower cost per training run.
- **NPU performance.** For a global accelerator manufacturer, benchmarked ResNet-50 (25M, FP16/INT8) and TinyLlama (1.1B, FP16/INT8) inference across Versal VE2308 FPGA and AMD Ryzen 7040 XDNA. Implemented C++20 operator tiling for convolution, multi-head attention, and normalization, delivering 2.6x faster Transformer inference, 1.3x faster CNN inference, ∼60% less off-chip DRAM traffic, and ∼2x better performance per watt while achieving interactive sub-50 ms token latency constraint.
- **Inference at scale.** For a top-tier cloud infrastructure provider, stabilized production vLLM and SGLang serving for MiniMax-M2.5 (230B, FP8) on AMD Instinct MI355X with ROCm 7.0. Diagnosed and patched an aiter 0.1.2 memory-layout indexing defect in the FlashAttention backend, triggered by FP8 KV-cache shuffling with a block size of 32, that caused incorrect memory allocation and worker crashes. The fix increased concurrent-request capacity by 55%, sustained over 40,000 tokens/sec per 8-GPU node, and reversed a 3x p99 time-to-first-token regression.
- **Inference benchmarking infrastructure.** Built an Python/Kubernetes CLI tooling that provisions cloud GPUs and runs controlled sweeps across CUDA/HIP Graph execution, KV-cache settings and block sizes, quantization, and vLLM scheduler policies. Added live performance visualization and reproducible diagnostic reports, enabling a ∼42% increase in best measured P90 per-user throughput, reducing regression-debugging time by 80%, and saving 15+ engineering hours per week.
- **Technical leadership.** Led optimization engagements with three enterprise customers, defining success metrics and presenting trade-offs to engineering teams and leadership. Onboarded seven engineers.

Silo AI **Helsinki, Finland**
AI R&D lab for enterprise clients
Senior AI Engineer *Nov 2023 – Aug 2024*

- **LLM evaluation.** Built a system that distinguishes AI-written code from human-written code, using LLM-based generation and evaluation pipelines. The first deployed version reached ∼**90%** precision at **80%** recall and cut manual review by ∼**60%**.
- **Service ownership and fine-tuning.** Owned the avatar-generation service end to end, from system design and model integration through deployment and production operation. Fine-tuned diffusion models with LoRA and combined subject- and style-specific adapters to preserve identity without full-model retraining.
- **RAG search.** Delivered patent search for a global telecom company, combining vector and keyword search with LLM re-ranking. Prior art searches went from hours to minutes, with quality measured against expert relevance judgments.

Skoltech **Moscow, Russia**
AI R&D lab for enterprise clients
Software/Research Engineer *Jul 2019 – Apr 2023*

- Built a reinforcement learning pipeline with self-play and opponent modeling for a game AI client. It matched an expert rule-based system and replaced hand-written rules.
- Built a real-time 3D gesture recognition system for a global telecommunications company by compressing spatial-temporal transformer models to run on low-power edge devices.

FusionSoft **Lipetsk, Russia**
Software development studio
Software Engineer *Dec 2017 – Mar 2019*

- Built containers and GPU training job scheduling in Docker/Kubernetes, Helm charts deployments, and Prometheus/Grafana monitoring in a custom MLOps platform.

Lipetsk State Technical University **Lipetsk, Russia**
Technical university and applied mathematics department
Assistant / Associate Professor *Feb 2010 – Dec 2017*

- Taught machine learning, optimization, numerical methods, and probability theory; supervised eight students and delivered applied ML/control prototypes, including robot localization and control.

EDUCATION

Voronezh State Technical University **Voronezh, Russia**
Ph.D. in Applied Mathematics *2010 – 2014*
Thesis: Numerical methods for modeling and control of multi-agent systems.

Lipetsk State Technical University **Lipetsk, Russia**
M.Sc. in Applied Mathematics *2005 – 2010*
Thesis: Idempotent mathematics methods for robotic manipulator control.

PUBLICATIONS

1. Nikolaev D. et al. Effect of Video Game Players' Emotions on Team Performance in the Course of eSports Tournament. *IEEE Internet of Things Journal*, 2020.
2. Nikolayev D. et al. Tropical Neighborhood and Neural Models for Modeling of Greedy Organizational Systems. *Global Journal of Pure and Applied Mathematics*, 2016, pp. 4741–4747.
3. Nikolayev D.A. Idempotent algebra methods for modelling of hierarchical multiagent systems motion. In G.L. Litvinov, V.P. Maslov, A.G. Kushner, S.N. Sergeev (Eds.), *Tropical and Idempotent Mathematics*. Moscow, 2012, pp. 193–197.

ADDITIONAL INFORMATION

Location: Helsinki, Finland (EU Blue Card; mobile within EU; open to remote and relocation)
Languages: Russian (native), English (full working proficiency), Finnish and German (conversational)